Θάλπος · warmth, comfort

Thalpos

A smart room monitor that knows when someone is in, and tells you only what matters.



Charalampos Mentsios · ESP8266 + FastAPI + React

Where the idea came from

- Started as a learning project: one cheap sensor on a breadboard
- First goal: see live room data on a screen
- Each working version raised the next question: **what should the room do with this data?**
- The name: **Θάλπος**, Greek for warmth and comfort

August 2026

Smart Building AI OS

DHT11, MQTT, FastAPI and a Streamlit page

Next

AIOS v6

React, 3D home model, forecasts, Telegram and Alexa

Today · v6.20

Thalpos

SHT31, mmWave radar, presence-aware alerts

The problem

We notice a room’s air only when it already feels wrong.

Humidity creeps up unseen

- Window shut overnight, humidity climbs
- Condensation, stale air, then mold

Airing is guesswork

- Is the air outside better or worse?
- How long has someone been inside?

Alerts that cry wolf

- Fire even when nobody is home
- Repeat on every flicker, so get muted

What Thalpos aims for

Measure what is really there

- Precise temperature and humidity
- Presence from a 24 GHz radar

Explain, don’t just display

- Health score, dew point, mold risk
- A 10-minute forecast and a next step

Speak up at the right moment

- Alerts only when someone can act
- Quiet hours, pause, no repeats

Local, private, low-cost

- Hobby-grade parts, one breadboard
- Data stays at home, no cloud account

Method: small versions, tested in the real room

Every feature went through the same loop:

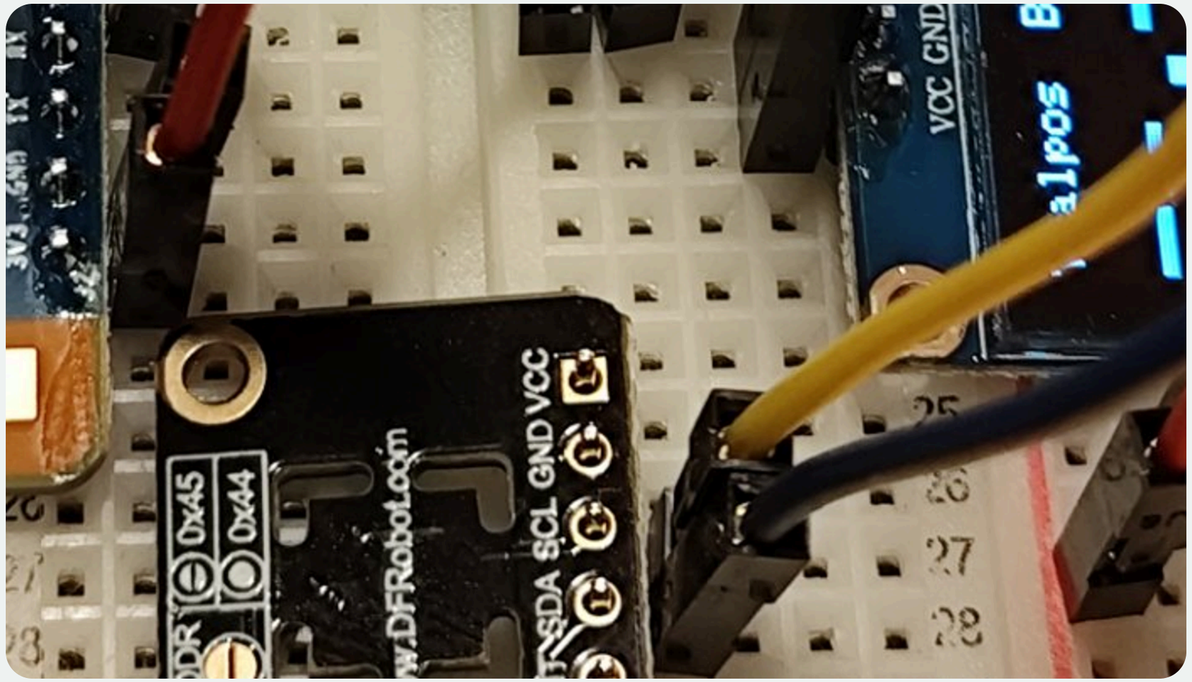
1. **Build one thin slice**: firmware to dashboard
2. **Try it on the real board**: walk in and out, open the window, read the logs
3. **Lock it in with tests**: each bug found becomes a pytest case
4. **Document and ship**: README, tagged commit, updated demo

83
automated backend tests, all passing

v6.20
reached one small release at a time

Online demo
Simulated sensor, same formulas as the backend

Hardware: one breadboard, all on 3.3 V



SHT31 and OLED share the I2C bus: SDA on D2, SCL on D1.

- **ESP8266 NodeMCU**: Wi-Fi, OTA updates
- **SHT31** (I2C 0x45): temperature, humidity
- **24 GHz mmWave radar**: presence on D5
- **SSD1306 OLED**: readings on the device
- **DHT11**: automatic fallback on D4
- Bus scan at boot shows what is connected

Architecture: from a pin to a phone



Zero configuration

- Backend broadcasts a beacon
- Board finds the broker itself

No cable for updates

- New firmware over Wi-Fi (OTA)
- Protected by a password

Private by default

- API listens only on the PC
- Phone connects via Tailscale

Code: presence that doesn’t flicker

- **A change must hold**: 5 s to enter, 60 s to leave
- **Every visit is logged** in SQLite: time in the room per day
- **Comfort alerts wait** while the room is empty
- **Critical levels and faults** always go through
- **Airing reminder** after 2 h if humidity climbs, once per stay

backend/presence.py

```
hold = ENTER_HOLD_SEC if present else LEAVE_HOLD_SEC
# 5 s to come in, 60 s to leave
if now - self._candidate_at < hold:
    return None  # a flicker, not a change

self._state = present
self._log("visit_started" if present
          else "visit_ended")
text = message(present, room, wall, stayed)
self._send(text, wall, self.on)
```

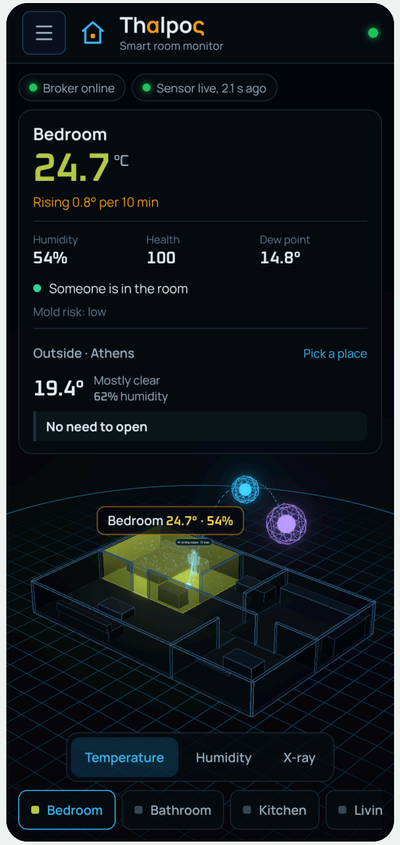
The dashboard: a living model of the home



- 3D twin built with three.js
- Rooms glow by temperature or humidity
- Hologram when someone is in
- Health score, dew point, mold risk
- Window advice from live weather
- Live packet stream

History, forecasts and the phone





- History: 1 hour to 30 days
- Time in the room per day
- 10-min forecast with confidence
- CSV export
- Installs on the phone

Every page of the dashboard

Seeing the room

- **Dashboard**: 3D twin with an X-ray view
- **Floorplan**: 2D overview of the rooms
- **Analytics**: history, time in the room
- **Predictions**: 10-minute forecasts
- **AI Brain**: rule-based comfort analysis

Acting on it

- **Alerts**: active alerts and their history
- **Reports**: statistics and CSV downloads
- **Settings**: limits, pause, quiet hours
- **MQTT Stream**: every packet as it arrives
- **Energy**: simulated, no meter yet

Notifications that earn their place

Telegram

- Alerts start and clear
- Someone enters or leaves
- Airing reminder
- Morning summary with chart
- /status, /today, /presence

Alexa

- Speaks new alerts only
- At most 5 a day
- Silent 23:00–08:00

IKEA lamp

- Lights up during an alert
- On for 10+ min, no flicker
- Local hub API, no cloud

Noise rules

- A change must hold 2 checks
- Same alert at most every 30 min
- Quiet hours and /mute

Secure and easy to run every day

Security

- MQTT needs a password, never anonymous
- Random passwords for backend and board
- API listens only on the PC itself
- Phone access through Tailscale
- Pre-commit check blocks keys and passwords

Running it

- Desktop icon starts backend, then dashboard
- Optional autostart with Windows, logs saved
- Separate badges: API, broker, sensor offline
- Telegram warns if the sensor gets stuck
- Board finds the broker after a restart

What I learned building it

Placement beats code

- Lying flat, the radar saw through the wall
- Upright and aimed at the door, it’s reliable

Touching isn’t connected

- The SHT31 was missing from the I2C scan
- Soldering the header pins fixed it

A filter beats a model

- Radar flicker caused false “came in” messages
- Simple hold times removed them

Fewer messages, more trust

- Comfort alerts held while nobody is in
- The bot became worth reading

What comes next

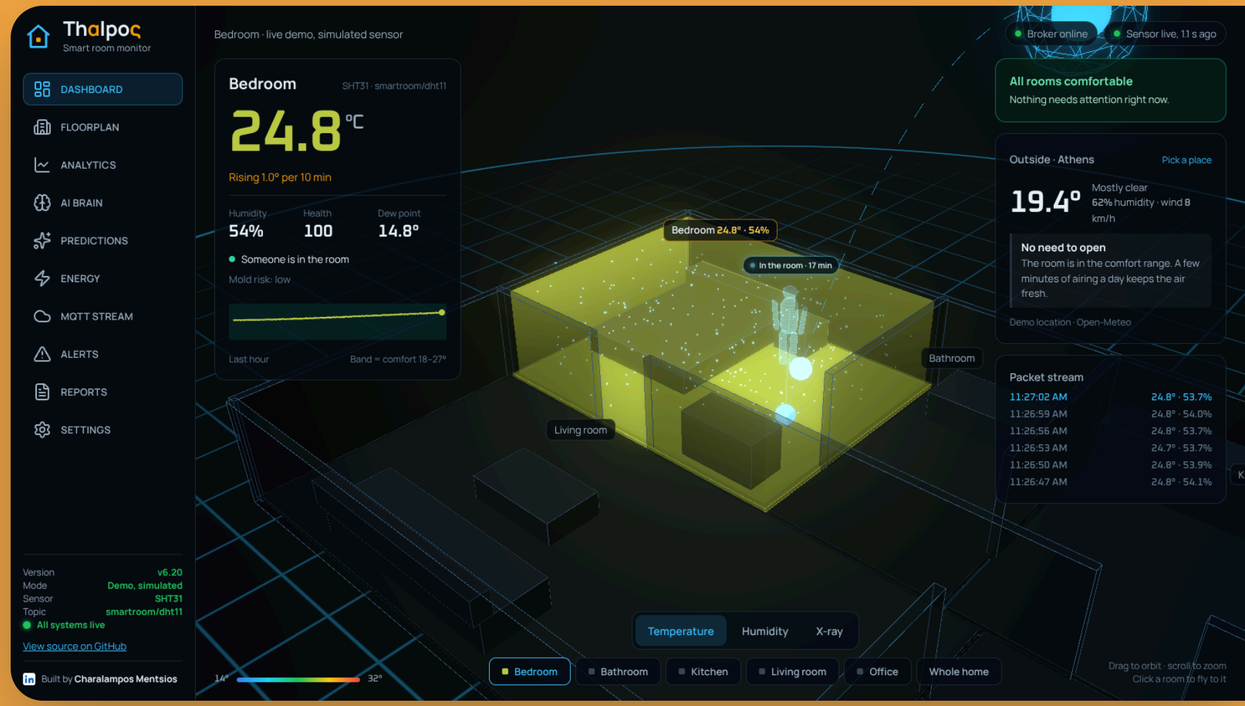
- **PM2.5 sensor**: air quality, not just comfort
- **LD2450 radar**: count people, not just detect
- **Air conditioner control**: cool only when someone is in
- **Raspberry Pi**: an always-on server instead of a laptop
- **Calibration offset**: match a reference thermometer

From monitoring to acting.

A new sensor is one more field in the MQTT message and one more engine in the backend.

Try it yourself

- **Live demo**: xarismnt18-bot.github.io/thalpos
- **Source**: github.com/xarismnt18-bot/thalpos
- No hardware needed for the demo



Charalampos Mentsios · thank you